

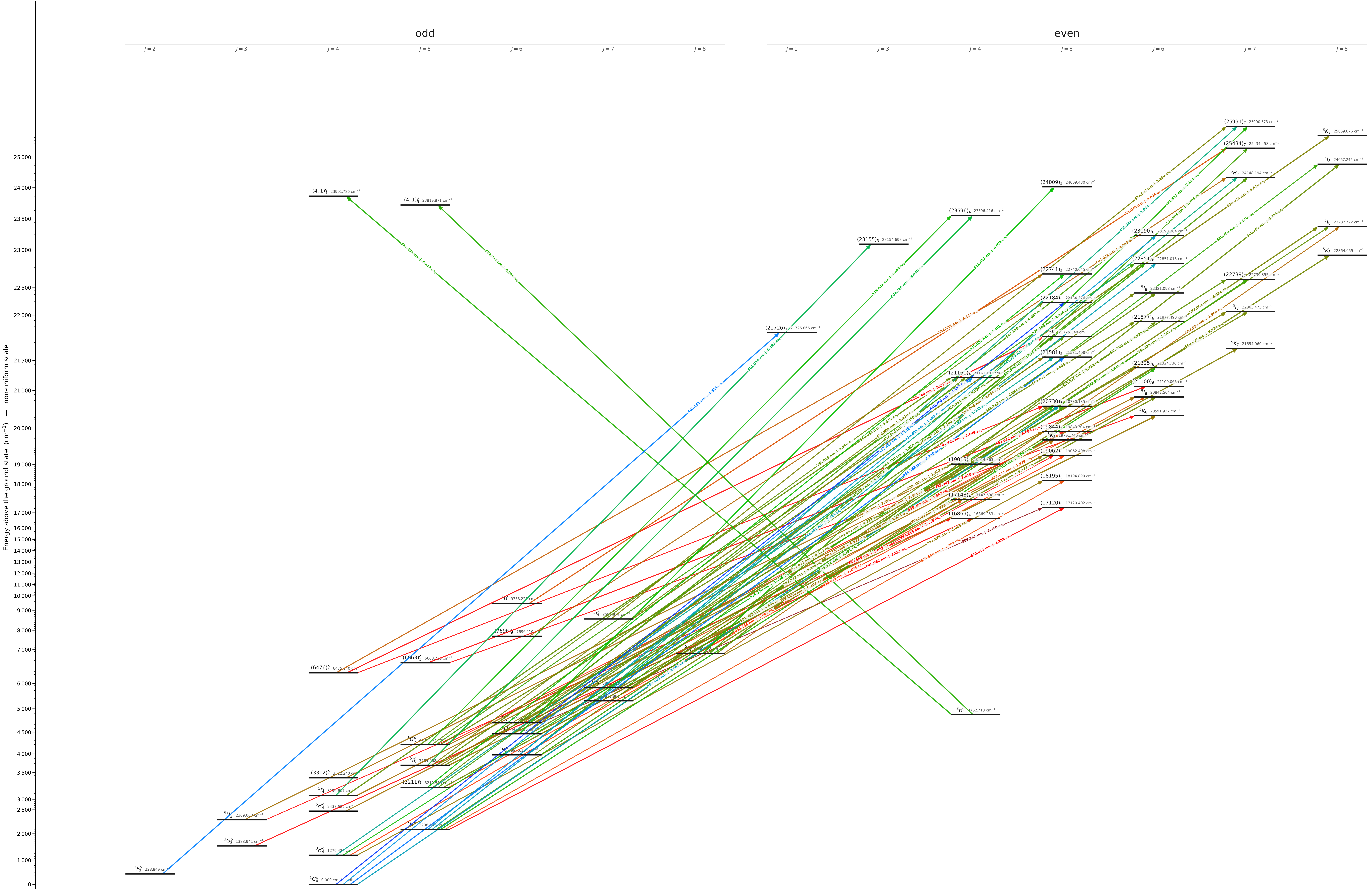
# Cerium

Ce (Ce II). Strongest classified transitions below 2 μm (NIST): 62 levels, 77 lines.

Ce <sup>+</sup> ionisation limit 44672.00 cm<sup>-1</sup> = 5.53862 eV (223.854 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $[U\|erU']$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory; ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



## Strongest transitions

| Transition                                                     | λ vac (nm) | λ air (nm) | ν (THz)  | [U erU'] (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use |
|----------------------------------------------------------------|------------|------------|----------|-----------------------------|----------------------|----------------|------------|-----|
| <sup>3</sup> f <sub>7</sub> - (23155) <sub>3</sub>             | 501.0501   | 500.9103   | 598.3284 | 5.191                       | 6.200e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - (23596) <sub>4</sub>             | 504.2252   | 504.0846   | 594.5606 | 5.000                       | 4.390e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - (24009) <sub>5</sub>             | 511.4128   | 511.2704   | 586.2044 | 4.976                       | 3.410e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - (25991) <sub>7</sub>             | 521.3372   | 521.1920   | 575.0453 | 7.111                       | 4.820e+07            | -              | -          | -   |
| <sup>3</sup> H <sub>4</sub> - (4, 1) <sub>2</sub> <sup>o</sup> | 522.4915   | 522.3460   | 573.7748 | 6.417                       | 6.500e+07            | -              | -          | -   |
| <sup>3</sup> H <sub>4</sub> - (4, 1) <sub>2</sub> <sup>o</sup> | 524.7374   | 524.5913   | 571.3191 | 6.206                       | 4.910e+07            | -              | -          | -   |
| <sup>3</sup> H <sub>5</sub> - <sup>3</sup> H <sub>7</sub>      | 545.0754   | 544.9239   | 550.0018 | 6.403                       | 3.420e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - (21161) <sub>4</sub>             | 556.6508   | 556.4962   | 538.5647 | 5.625                       | 4.130e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - <sup>3</sup> I <sub>5</sub>      | 556.7513   | 556.5968   | 538.4674 | 5.076                       | 2.750e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - <sup>3</sup> I <sub>6</sub>      | 560.2832   | 560.1277   | 535.0731 | 9.794                       | 6.500e+07            | -              | -          | -   |
| (3211) <sub>2</sub> <sup>o</sup> - <sup>3</sup> I <sub>6</sub> | 567.1532   | 566.9958   | 528.5917 | 6.373                       | 3.470e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - <sup>3</sup> K <sub>8</sub>      | 569.8573   | 569.6993   | 526.0834 | 8.934                       | 5.140e+07            | -              | -          | -   |
| <sup>3</sup> H <sub>5</sub> - <sup>3</sup> I <sub>6</sub>      | 572.0623   | 571.9036   | 524.0556 | 8.924                       | 5.070e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - <sup>3</sup> K <sub>6</sub>      | 578.9750   | 578.8145   | 517.7986 | 8.426                       | 4.360e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - <sup>3</sup> K <sub>7</sub>      | 581.4527   | 581.2915   | 515.5922 | 6.608                       | 3.000e+07            | -              | -          | -   |
| <sup>3</sup> f <sub>7</sub> - <sup>3</sup> K <sub>9</sub>      | 594.2502   | 594.0856   | 504.4886 | 6.107                       | 2.770e+07            | -              | -          | -   |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).